

Cisco Router & Switch

Password Recovery and ROMMON IOS Recovery Lab

Purpose: Practice recovering administrator access to Cisco IOS devices when the configured password is unknown, and practice recovering/uploading an IOS image from ROMMON. Perform these procedures only on equipment or lab devices you are authorized to administer.

1. Learning Objectives

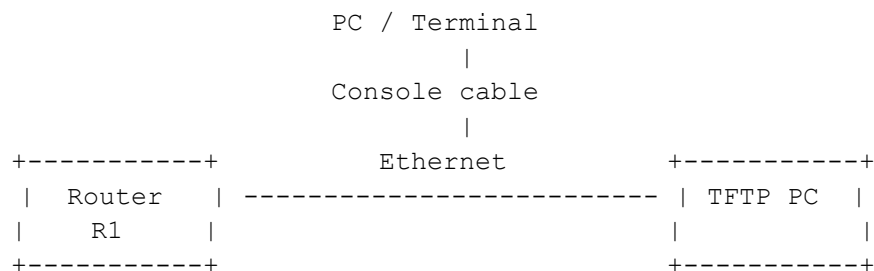
- Recognize the difference between normal IOS mode, boot loader/ROMMON mode, and configuration-register recovery.
- Recover access to a Cisco router when the enable password is unknown.
- Recover access to a Cisco switch using the platform's supported password-recovery procedure.
- Understand how configuration-register settings affect router boot behavior.
- Back up the existing configuration before completing password recovery.
- Identify IOS recovery methods such as TFTP and XMODEM and understand when each is appropriate.
- Verify IOS image integrity and confirm the device boots normally after recovery.

2. Important Platform Note

Cisco recovery commands are platform and software-version dependent. Modern Catalyst switches, older Catalyst switches, ISR routers, and newer IOS XE platforms do not all use the same ROMMON commands. The commands below are organized as common lab procedures, but the exact command set must be checked against the specific Cisco model and software release.

Do not erase startup-config or format flash unless the lab objective specifically requires it. For password recovery, the normal objective is to preserve the existing configuration.

3. Lab Topology



For switch practice: PC/Terminal ---- Console ---- SW1

4. Suggested Lab Devices

Device	Purpose	Examples
Router	Password recovery + ROMMON IOS recovery	ISR/IOS router supported by your lab
Switch	Password recovery	Catalyst switch with console access
PC	Console terminal and optional TFTP server	Windows/Linux
Console cable	Access boot/ROMMON output	USB console / RJ-45 console as

5. Before Starting: Record Device Information

On a normally functioning device, collect this information before intentionally testing recovery:

- show version
- show running-config
- show startup-config
- dir flash:
- show file systems

Record the exact platform/model, IOS or IOS XE version, configuration-register value, flash contents, and current management IP address. These details are useful during recovery.

6. Router Password Recovery – Common IOS Procedure

6.1 Enter ROMMON

1. Connect to the router console port using a terminal application.
2. Reload or power-cycle the router.
3. During the boot process, send the platform-specific break sequence to interrupt normal boot.
4. Confirm that the device reaches a ROMMON prompt such as:

```
rommon 1 >
```

The exact break method varies by terminal software and platform. If the break sequence does not work, check the Cisco documentation for the specific router.

6.2 Change the Configuration Register for Recovery

A common classic-IOS router recovery setting is:

```
rommon 1 > confreg 0x2142
```

This tells many classic IOS routers to ignore startup-config during the next boot. It does NOT erase the configuration.

Then reset/reload the router using the command supported by the platform, commonly:

```
rommon 2 > reset
```

6.3 Preserve the Existing Configuration

After the router boots, it may ask whether you want to enter the initial configuration dialog. Answer No for this recovery lab. The router should boot without applying the old startup configuration.

```
enable
```

```
copy startup-config running-config
```

The second command merges the saved configuration into RAM while preserving it. Do not use 'erase startup-config' for this password-recovery exercise.

6.4 Set a New Password

```
configure terminal
```

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```
enable secret NewStrongEnablePassword
```

```
username admin privilege 15 secret NewStrongLocalPassword
```

If the device uses console/VTY authentication, review those lines as well and make sure the new local account or AAA method will allow authorized management access.

6.5 Restore Normal Boot Behavior

```
configure terminal
```

```
config-register 0x2102
```

```
end
```

```
copy running-config startup-config
```

```
show version
```

The common 0x2102 value is used on many classic Cisco routers for normal boot behavior. Verify the platform documentation before applying it.

7. Router Recovery Verification

```
show version
```

```
show running-config | include username
```

```
show running-config | include enable secret
```

```
show startup-config
```

Reload the router only after confirming the correct configuration-register setting and saving the configuration.

8. Catalyst Switch Password Recovery

Switch password recovery is model dependent. Many classic Catalyst switches use a boot loader prompt and allow the startup configuration to be bypassed or renamed so that the switch can boot without applying the unknown password.

8.1 Enter the Switch Boot Loader

1. Connect to the console port.
2. Power-cycle the switch.
3. Interrupt the boot process using the platform's documented break/reset procedure.
4. Look for a prompt similar to:

```
switch:
```

The exact prompt and available commands differ by Catalyst generation.

8.2 Check Flash and Preserve Configuration

```
dir flash:
```

Look for the configuration file and IOS image. On many classic Catalyst platforms, the startup configuration is stored in a file such as config.text. Some platforms also use vlan.dat.

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A common classic recovery technique is to rename the configuration file rather than delete it. For example, on platforms where the boot loader supports it:

```
rename flash:config.text flash:config.text.old  
  
boot flash:<IOS-image-name.bin>
```

Do not blindly enter these commands on every Catalyst model. First use 'help' or '?' at the boot loader and verify that the syntax is supported.

8.3 After the Switch Boots

Answer No to the initial configuration dialog if presented. Then enter privileged EXEC mode and restore the saved configuration using the platform-appropriate method.

```
enable  
  
copy flash:config.text.old running-config
```

If the filename or boot loader syntax differs, use the actual file name shown by 'dir flash:'.

8.4 Set New Credentials and Save

```
configure terminal  
  
enable secret NewStrongEnablePassword  
  
username admin privilege 15 secret NewStrongLocalPassword  
  
end  
  
copy running-config startup-config
```

9. IOS Image Recovery / Upload from ROMMON

If the IOS image is missing, damaged, or the device cannot boot normally, recovery may be possible from ROMMON/boot loader. The method depends heavily on the platform.

9.1 Method A – TFTP Recovery

TFTP is generally preferable to XMODEM when the platform supports a network-based recovery command, because it is significantly faster.

Prepare the TFTP server:

- Install/enable a TFTP server on the authorized lab PC.
- Place the correct Cisco IOS image in the TFTP root directory.
- Temporarily configure the PC's Ethernet interface with an IP address on the same network as the router's recovery interface.
- Connect the router recovery Ethernet interface directly to the PC or through the lab switch.
- Disable or permit the required firewall/TFTP traffic for the lab.

At ROMMON, inspect the available commands:

```
rommon 1 > ?
```

If the platform supports a TFTP recovery command such as 'tftpdnld', the command and required environment variables will be displayed by the platform's help output.

A classic IOS router example is conceptually:

```
rommon 1 > IP_ADDRESS=192.168.10.1  
rommon 2 > IP_SUBNET_MASK=255.255.255.0  
rommon 3 > DEFAULT_GATEWAY=192.168.10.254  
rommon 4 > TFTP_SERVER=192.168.10.10  
rommon 5 > TFTP_FILE=c1900-universalk9-mz.SPA.x.x.bin  
rommon 6 > tftpdnld
```

The variable names and commands are examples for platforms that support this ROMMON workflow. Do not assume that a different router supports the same syntax. Verify with '?' and the Cisco documentation for the exact model.

9.2 Method B – XMODEM / Console Recovery

Some Cisco platforms support transferring an IOS image over the console connection using XMODEM from ROMMON or boot loader. This is much slower than Ethernet/TFTP and should normally be treated as a recovery method of last resort.

Typical workflow:

1. Confirm that the boot loader/ROMMON provides an XMODEM receive command.
2. Connect the console at the speed required by the platform; some platforms allow a higher baud rate for recovery.
3. Start the XMODEM receive command on the Cisco device.
4. Use the terminal application's file-transfer function to send the IOS .bin file with XMODEM.
5. Wait for the transfer and flash write to finish; do not power off the device.
6. Verify the image in flash and boot it using the platform's supported boot command.

Example syntax on some platforms may look like 'xmodem -c <filename>'. The exact command varies; use '?' before attempting the transfer.

10. IOS Image Verification

After copying an IOS image, verify that the file exists and is the expected size:

```
dir flash:  
  
verify /md5 flash:<IOS-image-name.bin>
```

Use the MD5/SHA hash published by Cisco for the exact image when available. A mismatched hash can indicate a corrupted or incorrect file.

11. Booting a Recovered IOS Image

The boot syntax varies by platform. A common IOS router example is:

```
boot flash:<IOS-image-name.bin>
```

Once IOS is running, configure the correct boot statement if supported, verify the configuration register, and save the configuration.

12. Password Recovery vs. IOS Recovery

Problem	Primary Goal	Typical Environment	Important Action
Unknown enable password	Regain administrative access	IOS already boots	Preserve and merge startup-config
Unknown switch password	Regain administrative access	Boot loader available	Use platform-specific password recovery
IOS image missing/corrupt	Restore bootable IOS	ROMMON/boot loader	Use supported TFTP/XMODEM/USB method
Wrong boot variable	Boot correct image	ROMMON/IOS	Inspect flash and boot variables

13. Troubleshooting Checklist

Symptom	Check
No ROMMON/boot-loader prompt	Confirm console connection, terminal settings, and the correct platform-specific break procedure.
Unknown command in ROMMON	Use '?' or 'help'. ROMMON commands vary significantly by platform.
Router boots into setup dialog	Do not erase configuration. If password recovery was intended, follow the configuration-register recovery workflow.
Startup configuration disappeared	Stop and verify whether it was renamed/ignored rather than deleted. Check flash/NVRAM carefully.
TFTP transfer fails	Check physical link, PC IP, server IP, firewall, TFTP root, filename, and ROMMON-supported variables.
XMODEM is extremely slow	This is expected compared with TFTP. Confirm the correct console speed and transfer protocol.
IOS image will not boot	Check image/platform compatibility, file integrity/hash, flash space, and boot syntax.
Device repeatedly enters ROMMON	Check boot variable, image filename, flash contents, and configuration-register setting.